

Mechanics Of Materials

- MECHANICS OF MATERIALS
- LOADS
- STRESS
- STRAIN



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Dr. Kashif Saeed (kashif.saeed@nu.edu.pk)

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Mechanics Of Materials

Basic Mech. Engg.

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Loads

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Stress

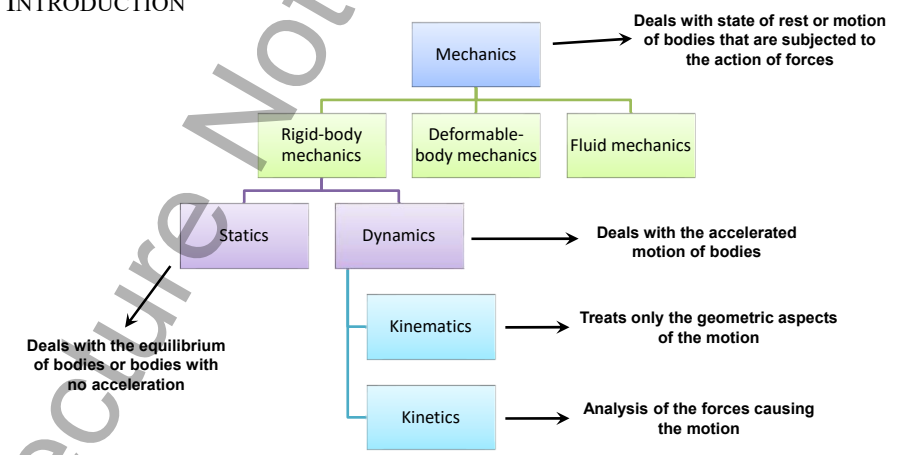
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INTRODUCTION



Deals with state of rest or motion of bodies that are subjected to the action of forces

Deals with the equilibrium of bodies or bodies with no acceleration

Deals with the accelerated motion of bodies

Treats only the geometric aspects of the motion

Analysis of the forces causing the motion

Essential for the design and analysis of many types of structural members, mechanical components, or electrical devices encountered in engineering

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INTRODUCTION





Fig.1 Flat Face Grips

Fig.2 Test Pieces

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APPLICATIONS





Millau Viaduct

Eiffel Tower

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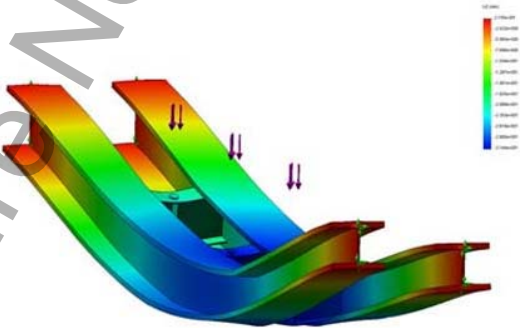
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DESIGN



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TESTING

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MISTAKES



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Loads

EXTERNAL LOADS

External Loads

Surface Forces

Body Forces

Concentrated Load

Distributed Load

Acts on a Point (Negligible Area)

Acts on a Line (Narrow Strip)

Surface forces are caused by the direct contact of one body with the surface of another

Body force is developed when one body exerts a force on another body without direct physical contact

Idealizations

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Sponge Model ^[1]



Concentrated Load



Distributed Load

[1] – Website: http://www.mace.manchester.ac.uk/project/teaching/civil/structuralconcepts/Statics/stress/stress_mod3.php Visited On: 10-05-2015

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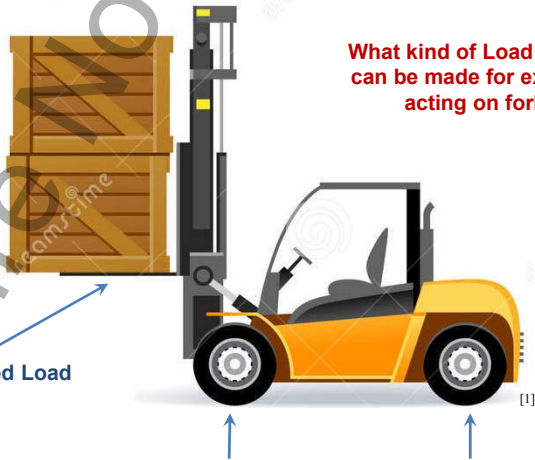
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Distributed Load

Concentrated Load

What kind of Load Idealizations can be made for external loads acting on fork lifter?

[1] – Website: <http://www.dreamstime.com/royalty-free-stock-photography-forklift-image25095357> Visited On: 13-05-2015

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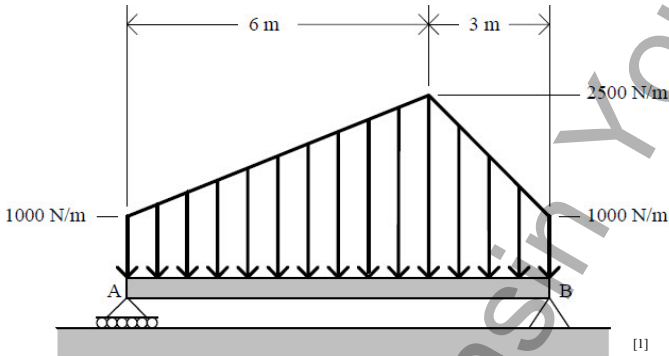
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Loads

EXTERNAL LOADS - DISTRIBUTED LOADS



[1]

Resultant force is equivalent to the area under the distributed loading curve

Resultant acts through the centroid or geometric center of this area

[1] – Website: <http://www.chegg.com/homework-help/questions-and-answers/beam-shown-bears-distributed-load-indicated-determine-reactions-supports-b-q873787>

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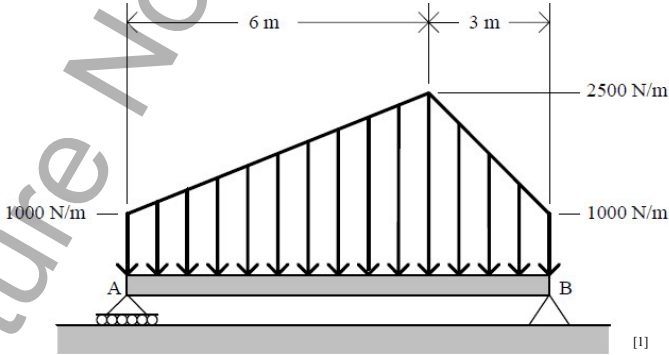
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Loads

EXTERNAL LOADS - DISTRIBUTED LOADS



[1]

What is the Resultant Force Acting of the Bar?

[1] – Website: <http://www.chegg.com/homework-help/questions-and-answers/beam-shown-bears-distributed-load-indicated-determine-reactions-supports-b-q873787>

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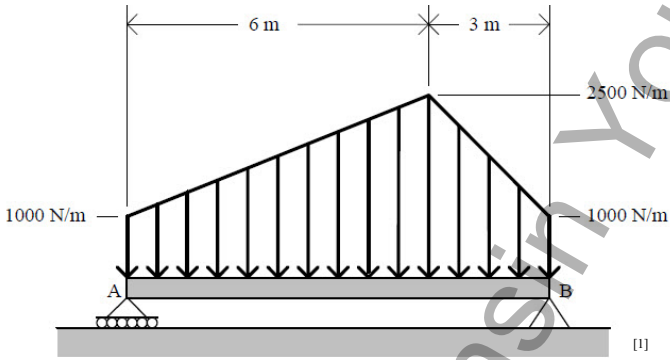
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Loads

EXTERNAL LOADS - DISTRIBUTED LOADS



[1]

Where is this Resultant Acting?

[1] – Website: <http://www.chegg.com/homework-help/questions-and-answers/beam-shown-bears-distributed-load-indicated-determine-reactions-supports-b-q873787>

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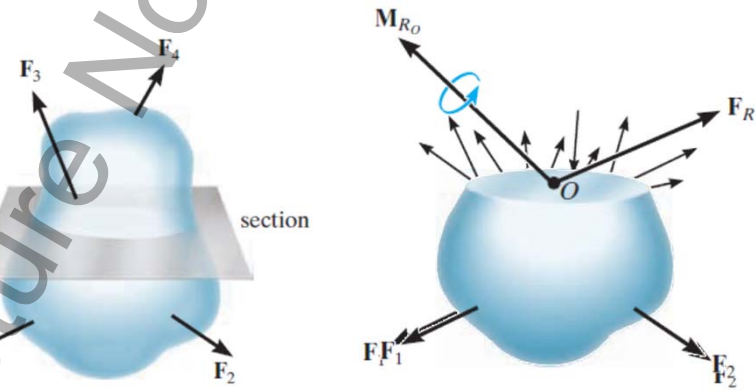
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Loads

INTERNAL RESULTANT LOADING



[1]

[1] - Russell C. Hibbeler. Mechanics of Materials 8th edition, Fig 1-2 (a,b,c), pp. 7

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Logo

Loads

INTERNAL RESULTANT LOADING

Normal Force (N) acts perpendicular to area

Shear Force (V) acts Parallel to area

[1]

[1] - Russell C. Hibbeler. Mechanics of Materials 8th edition, Fig 1-2 (d), pp. 8

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INTERNAL RESULTANT LOADING

Tension

Compression

Normal Force (N) acts perpendicular to area

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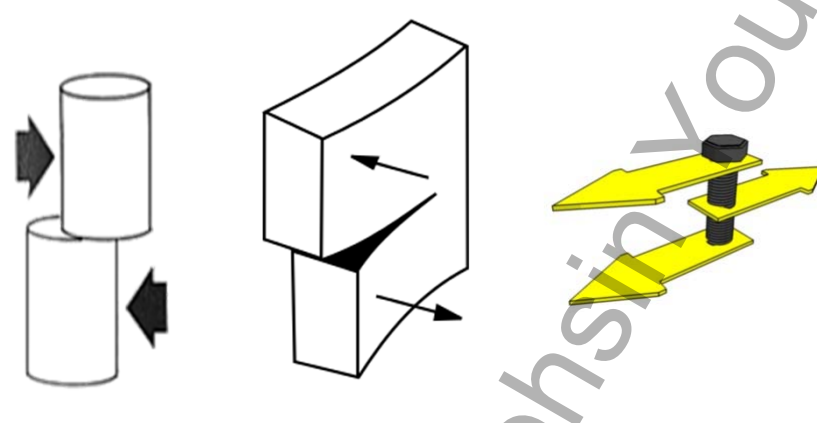
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INTERNAL RESULTANT LOADING



Shear Force (V) acts Parallel to area

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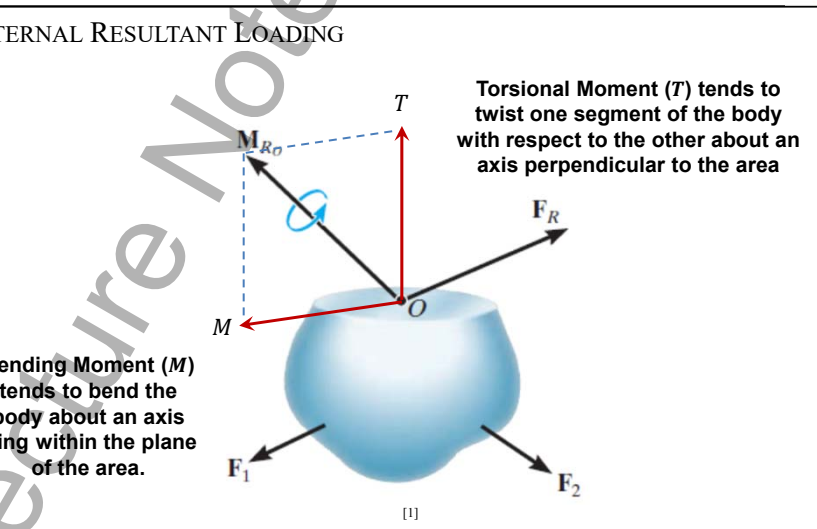
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INTERNAL RESULTANT LOADING



Torsional Moment (T) tends to twist one segment of the body with respect to the other about an axis perpendicular to the area

Bending Moment (M) tends to bend the body about an axis lying within the plane of the area.

[1]

[1] - Russell C. Hibbeler. Mechanics of Materials 8th edition, Fig 1-2 (d), pp. 8NU - FASTDr. Kashif Saeed (kashif.saeed@nu.edu.pk)ME101 (Spring 2017)19 / 43

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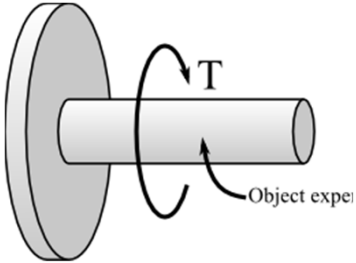
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INTERNAL RESULTANT LOADING



Torsional Moment

Torsional Moment (T) tends to twist one segment of the body with respect to the other about an axis perpendicular to the area



Bending Moment

Bending Moment (M) tends to bend the body about an axis lying within the plane of the area.

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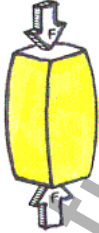
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Loads

INTERNAL RESULTANT LOADING

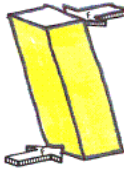
Identify Internal Loading



Compression



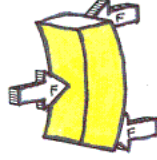
Tension



Shear



Torsion



Bending

[1] – Website: <http://legacy.mos.org/discoverycenter/aotm/2012/04> Visited On: 01/05/2015

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COPLANAR LOADINGS

The diagram shows a beam element of length Δx with a central point O . A vertical y -axis and a horizontal x -axis are centered at O . Four external forces are applied: F_1 and F_2 on the left end, and F_3 and F_4 on the right end. Internal forces at the right end are labeled: V (Shear Force) acting vertically upwards, N (Normal Force) acting horizontally to the right, and M_o (Bending Moment) acting counter-clockwise. A label 'y section' is placed near the y -axis.

How Can One Calculate V, N and M_o ?

$\sum \vec{F} = 0 \longrightarrow N \text{ and } V$

$\sum \vec{M}_O = 0 \longrightarrow M$

[1] - Russell C. Hibbeler. Mechanics of Materials 8th edition, Fig 1-3, pp. 9

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